

Muhammad Hilmy Putra Risma

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SUMMARY

Motivated and detail-oriented Mathematics graduate with a deep focus on data analytics and nearly three years of dedicated experience through academic projects and certified coursework. Skilled in data processing, visualization, and statistical modeling using Python, SQL, Excel, and Tableau. Passionate about the integration of mathematics, IT, and business to drive impactful, data-driven decisions and contribute meaningfully to real-world problem-solving.

SKILLS

- Data Analysis & Machine Learning:** Python with libraries like SciPy, Pandas, Matplotlib, Scikit-Learn, TensorFlow, Keras.
- Data Management & Querying:** Excel, PostgreSQL, MySQL, Pentaho, Talend, DBeaver, Google BigQuery, Airflow.
- Data Visualization & BI Tools:** Tableau, Metabase, Looker Studio, PowerBI.
- AWS Platform:** S3, Glue, Redshift, Athena, SageMaker, QuickSight, Kinesis, Step Functions, Lambda.

EXPERIENCE

- Data Analytics | PT Bank Mandiri (Persero) Tbk**

Jul 2026 – Present

 - Workforce Analytics & Visualization:** Analyzed and visualized employee demographic data using Python, Excel, and Power BI to provide insights into workforce composition and support Human Capital monitoring.
 - Employee Data Management:** Routinely updated and maintained employee datasets using Python automation, ensuring data accuracy, consistency, and timely availability for Human Capital reporting and analysis.
- Data Analytics | PT Bank Rakyat Indonesia (Persero) Tbk · Internship**

Dec 2025 – Jun 2026

 - Performance Analysis:** Conducted employee productivity analysis across 18 Regional Offices by examining key performance indicators and demographic factors to identify workforce performance patterns and trends using Python, Excel, and Tableau.
 - Dashboard Development:** Developed productivity and employee profiling dashboards based on data from 26,000+ employees using Tableau, Excel, and Python, supporting performance monitoring and management decision-making.
- Data Engineer | AnterAja · Internship**

Jul – Dec 2025

 - Data Management & Querying:** Managed and explored large-scale datasets using PostgreSQL and DBeaver, ensuring accurate data extraction and efficient query execution for business reporting needs.
 - ETL Development:** Built and automated ETL pipelines using Pentaho Data Integration and Talend to clean, transform, and load raw data into structured formats across systems.
 - Dashboard Development:** Created interactive dashboards in Metabase to monitor Job Orders, Cost, and Unit Utilization, providing real-time insights to support operational and strategic decisions.
- Business Database Analyst | PT Pegadaian · Internship**

Aug – Dec 2023

 - Data Visualization:** Developed interactive dashboards using Tableau and Excel, including a geographic potential map by analyzing customer distribution across 7 regions with population data, and a daily-updated KPI monitoring dashboard for 64 outlets to help branch managers track performance trends.
 - Customer Churn Analysis:** Conducted customer churn analysis using independent sample t-tests to identify significant differences in customer characteristics and behavioral factors associated with churn, providing insights to support customer retention strategies.
 - Predictive Analysis:** Conducted a three-year analysis on Outstanding Loan (OSL) data and performed forecasting using the ARIMA time series model, achieving a forecast accuracy of 97%.

EDUCATION

- UNIVERSITY OF INDONESIA**

Aug 2020 – Jan 2025

B.Sc. in Mathematics (Specialization: Data Sciences)
Emphasized Data Analysis, Statistical Modeling, Machine Learning, and Deep Learning

PROJECTS

- Comparative Evaluation of Multi-Large Language Model APIs for Sentiment Analysis**

Apr 2025

Developed a sentiment analysis system in Python integrating DeepSeek, Gemini, and GPT APIs to evaluate 2,400+ text samples; conducted multi-model classification analyzing accuracy, response time, and API cost; visualized results with confusion matrices achieving up to 99% accuracy.

Keywords: Sentiment Analysis, Natural Language Processing , Large Language Models, Machine Learning

Repository: github.com/MuhammadHilmyPutraRisma/multi-llm-sentiment
- Comparison of In-Context Learning in GPT for Sentiment Analysis Level Topic**

Jan 2025

Performed topic-level sentiment analysis on Tokocrypto app reviews by scraping data from the Play Store and App Store. Applied three in-context learning techniques using GPT models, with the best method achieving 99% accuracy. Utilized BERT for data representation and EFCM for topic clustering, identifying key topics with representative keywords for interpretation.

Keywords: Topic-Level Sentiment Analysis, In-Context Learning, GPT, BERT-EFCM

Repository: github.com/MuhammadHilmyPutraRisma/tokocrypto-sentiment
- Bitcoin Price Forecast with ARIMA Model**

Jan 2023

Developed ARIMA models in Python and R to forecast Bitcoin prices seven days ahead. Achieved high accuracy with an average RMSE of under \$200. Effectively visualized and presented insights and predictions to academic assessors, demonstrating strong forecasting and communication skills.

Keywords: Time Series Forecasting, Statistical Modeling, ARIMA Model, Predictive Analytics

Repository: github.com/MuhammadHilmyPutraRisma/bitcoin-arima-forecast

CERTIFICATES

- AWS Cloud Quest: Machine Learning** – AWS Training & Certification, Nov 2025
- AWS SimuLearn: Data Analytics** – AWS Training & Certification, Sep 2025
- Deep Learning Specialization** – DeepLearning.AI, May 2025
- Machine Learning Specialization** – Stanford University, May 2025
- Mathematics for Machine Learning** – Imperial College London, Apr 2025